

Ashwin K. Aggarwal

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EDUCATION

Cornell University – College of Engineering

Ithaca, NY

Bachelor of Science in Computer Science and Minor in Artificial Intelligence (GPA: 3.7)

Expected: May 2028

Relevant Coursework: Algorithms, Large-Scale Machine Learning, Deep Learning, Linear Algebra, Data Structures (Java), Probability and Stats, Computer Vision, Robotics, Computer Systems, Discrete Math, Functional Programming (OCaml)

EXPERIENCE

Virginia Commonwealth University

May 2026 – Aug 2026

Machine Learning Research Assistant, Wargo Lab

Richmond, VA

- Built ETL pipelines ingesting data from networked devices, automating Power BI dashboard generation; scheduled concurrent workflows to populate a SQL Server database, replacing manual data collection across 5+ client sites
- Extended a GPU-accelerated Monte Carlo simulation pipeline from 2D to 3D volumetric output on an HPC cluster, running end-to-end across 500+ simulated scans with CUDA; debugged critical object-generation error
- Co-authored a manuscript documenting the system design, automation workflow, and architecture for publication

Virginia Commonwealth University

May 2026 – Jun 2026

Data Engineering Intern, AI Department

Remote

- Designed Databricks pipeline for processing large datasets of unstructured, sensitive clinical text files from NoSQL database, sorting by medallion architecture to optimize RAG performance on downstream medical queries
- Leveraged OCR models to extract 100+ fields in a clinical note documents to ingest into agent for categorizing demographics and diagnosis; cut LLM API usage by 90% through prompt engineering and structured retrieval

Cornell Computational Biology Lab

Jan 2026 – May 2026

Undergraduate Machine Learning Researcher

Ithaca, NY

- Constructed and fine-tuned 3D CNNs (ResNet and U-Net architectures) for classifying parity of 500+ MRI scans; improved F1 Score (0.54 to 0.61) and overall accuracy using data augmentation, dropout, early stopping in PyTorch
- Created data pipelines normalize structural brain scans, iterating over 10+ model architectures to probe signal

Next Play

Jul 2025 – Aug 2025

Software Engineering Intern, AI/ML Team

Remote

- Built a real-time Computer Vision pipeline in TypeScript and React using MediaPipe pose detection, supporting rep counting and live feedback across 20+ fitness challenges, achieving 90% pose classification accuracy for users
- Developed and shipped 10+ backend features and REST API integrations for a sports game in Golang and Kotlin, deployed on AWS, contributing to gameplay functionality and cross-service communication at a fast-moving startup

PROJECTS

Multi-Agent Research Assistant | MCP, LangGraph, Docker, Claude API, Neo4j, Python

- Built a four-agent orchestration system on the Claude API with LangGraph that ingests, summarizes, and cross-links 30+ research papers into a Neo4j knowledge graph, surfacing 100+ connections keyword search missed
- Automated nightly paper ingestion with APScheduler and an arXiv MCP server; shipped a Streamlit interface with interactive pyvis graph visualization for tracing citation and concept paths across the corpus

AI Housing Lease Analyzer | LangChain, FAISS, BM25, NLP, Python, OpenAI API, Docker

- Built a hybrid RAG pipeline combining FAISS vector search and BM25 retrieval with LangChain and OpenAI to answer questions over long, multi-page contracts, reaching 94% faithfulness on RAGAS evaluation
- Launched a Streamlit app enabling natural language contract queries, cutting manual lease review workload by 80%

AI Basketball Shot Tracker | Python (OpenCV, PyTorch, Numpy, Matplotlib), Ultralytics (YOLO), RoboFlow

- Fine-tuned a YOLOv8 model on 300+ hand-labeled frames for ball and hoop detection, achieving 97% accuracy using RoboFlow for dataset annotation, augmentation, and tracking across dozens of model training and evaluation runs
- Classified made and missed shots using live ball and hoop position tracking via OpenCV, achieving 90%+ accuracy

SKILLS/INTERESTS

Skills: Java, Python, C#, C, C++, PyTorch, Docker, CI/CD, Kubernetes, CUDA, TypeScript, Go, Rust, AWS, ROS, MCP

Interests: Catan, Physical AI, Philosophy, Basketball, New England Patriots, Golf, Pickleball, Chess, Startups, Mariokart